Coding challenge

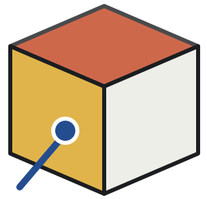
Photographs onto facades

From a phone photo to window sizes in metres

Adilzhan Yerzhan
20° GmbH, 16 September 2026

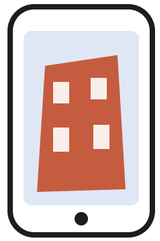
The challenge

The task



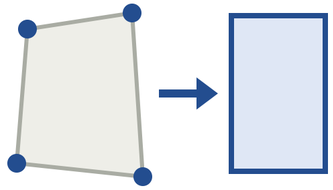
1

Pick a wall in the 3D model



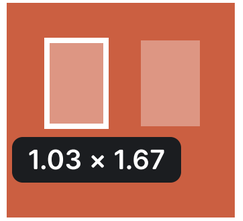
2

Give it a photo



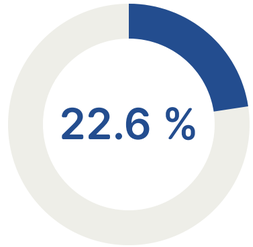
3

Straighten the photo onto the wall



4

Draw windows and doors in metres



5

Opening area and window-to-wall ratio

And say how accurate the numbers are, and how I know.

The challenge

Why it is hard



Perspective

A 14 mm ultra wide lens, verticals converge

Hidden parts

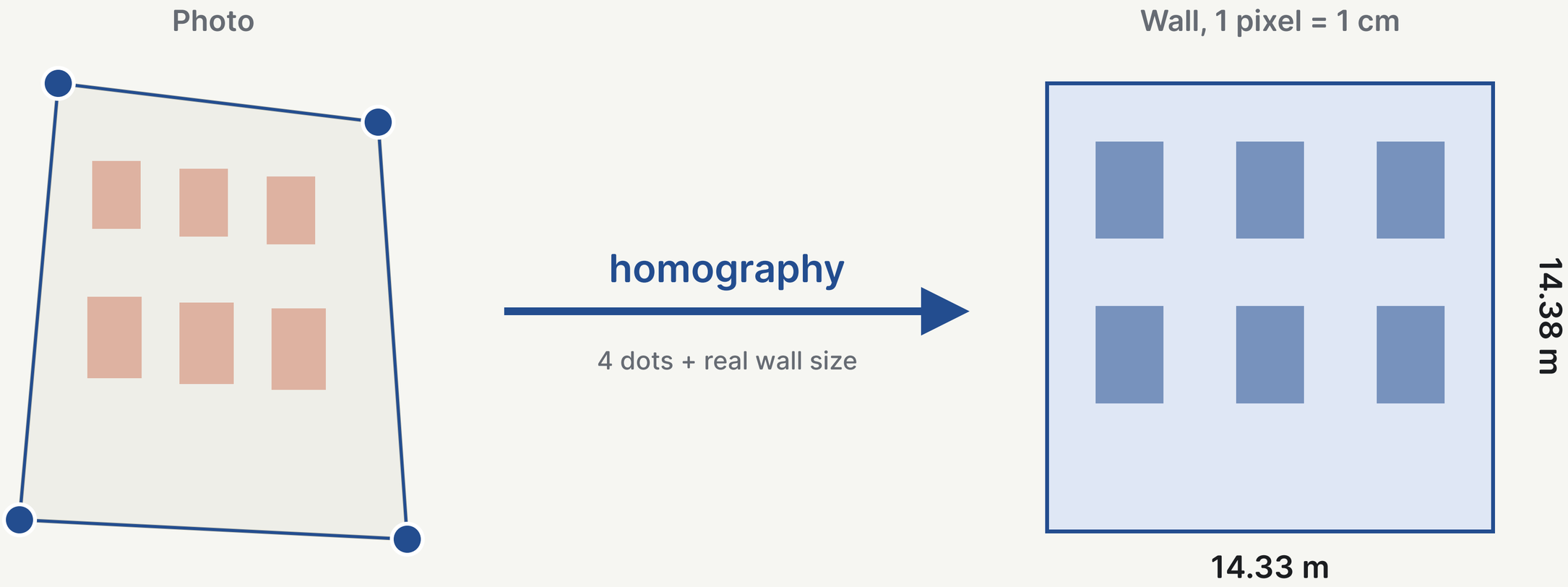
Cars, a skip, a tree, hard shadows

The model

Metric and georeferenced, but no basement and years old

The challenge

The idea in one picture



The model knows the real size of the wall. Four corners connect the photo to it.



Demo

BOo, then kY

Decisions and reasons

A person places the corners

Decided

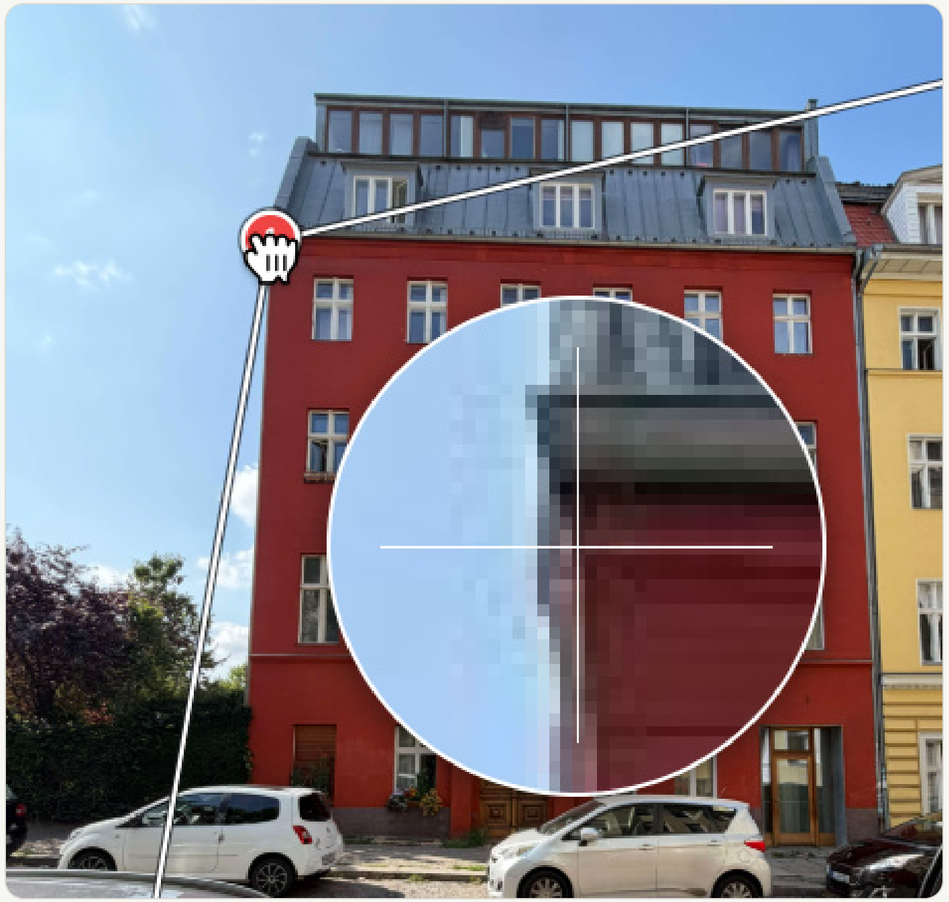
4 dots, dragged by a person, with a loupe

Why

Works on all three photos. A person sees the cars and the roof line. A wrong dot is easy to see and fix.

Rejected

Automatic corner detection. It fails where corners are hidden.



The loupe while I drag the top left corner

Decisions and reasons

My own homography, in lod2-core

Decided

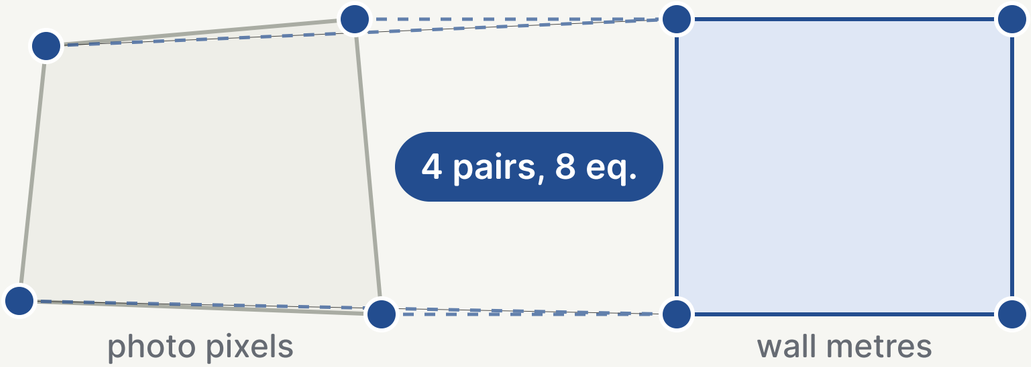
8 equations from 4 point pairs, solved in my own code

Why

Small maths, no new dependency, tested within 1 pixel, and I can explain every line

Rejected

OpenCV: a big dependency for one small solve




```
homographyFromPairs(from, to)
applyHomography(H, point)
invertHomography(H)
```

Tested

Known points land within 1 pixel, and a round trip comes back

Decisions and reasons

Openings drawn by hand, on the straight wall

Decided

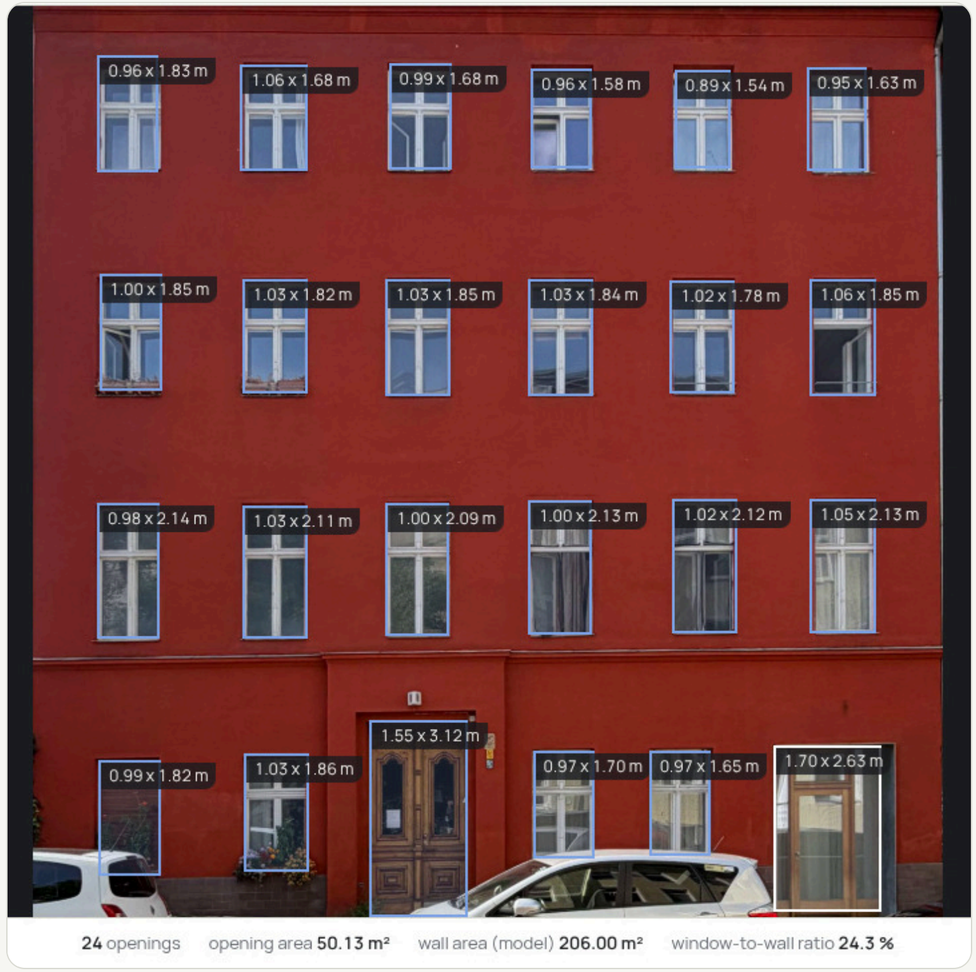
Rectangles drawn by a person, stored in wall metres

Why

Detectors fail behind cars, skips and trees. My boxes are the reference a detector is tested against.

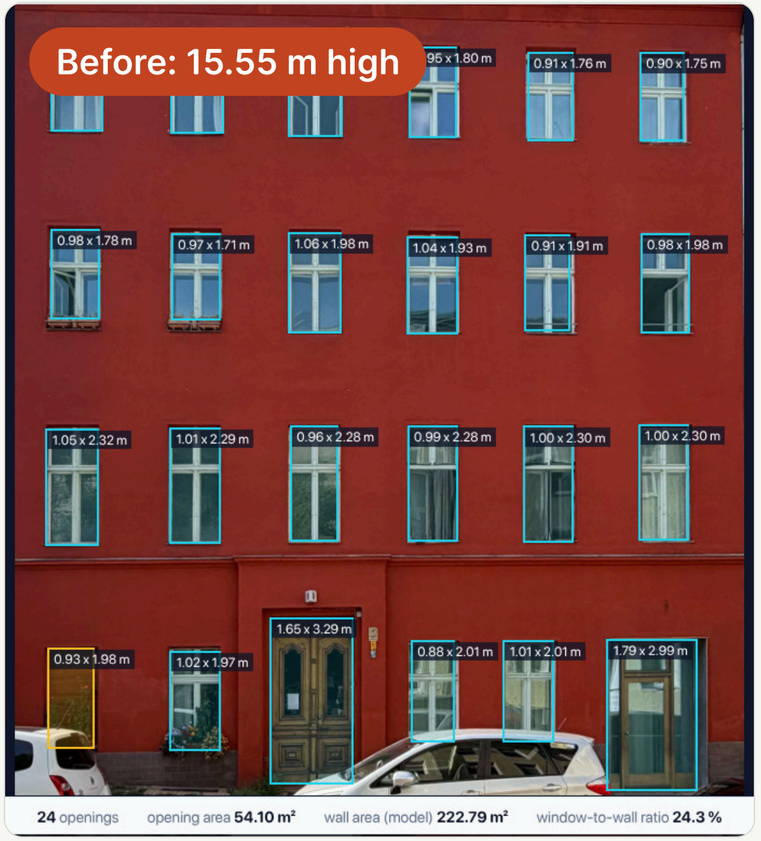
Rejected

Automatic detection first. It is my next step, on the rectified image.



Decisions and reasons

The key finding: measure above the ground





LoD2 has no basement, so the model wall goes below the street. On kY it went **10.94 m** down: 26.22 m became 15.28 m.

Decisions and reasons

How I know the numbers

Wall	Size above ground	Openings	Opening area	Ratio	
BOo	14.33 × 14.38 m	24	46.58 m²	22.6 %	
Atr	14.28 × 17.05 m	24	57.81 m²	23.7 %	
kY	9.00 × 15.28 m	12	24.82 m²	18.1 %, too low	

Drew every wall twice
Ratio moves 1 to 2 points

Identical windows
Differ by up to 16 to 17 cm

Doors
Up to 27 cm, bottoms behind cars

Decisions and reasons

Which wall is in the photo

Decided

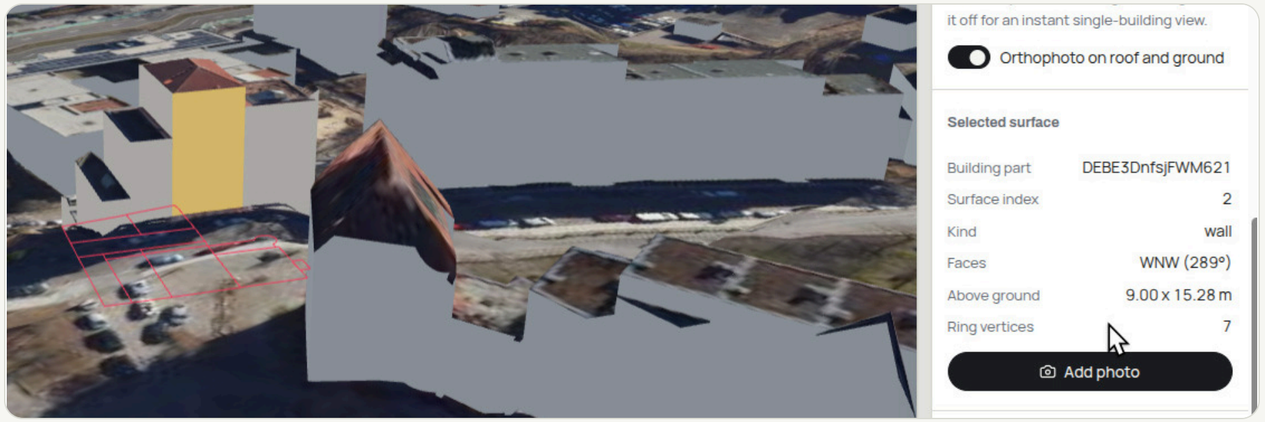
Show which way the wall faces, and warn when the photo's compass does not fit

Why

kY has 77 walls. A wrong wall makes every number wrong.

Rejected

The GPS position. The brief shows it is metres off.



kY middle wall, the panel says Faces WNW (289°). The brief says 289°.

The photo editor on kY

"Camera direction matches this wall (4 degrees off)."

optional

Pros and cons

What works well

Metric

Real metres from the model, not guessed

Checkable

Every corner and box can be seen and fixed by a person

Tested

All maths in lod2-core with node:test. Nothing in the UI measures.

Pros and cons

Where it is not accurate

Manual
24 boxes take a few minutes per wall

Hidden corners
BOo's bottom corners are behind cars

Only rectangles
A gable wall is not exact

kY ground floor
Not drawn, so the ratio is too low

No saving
A reload loses everything

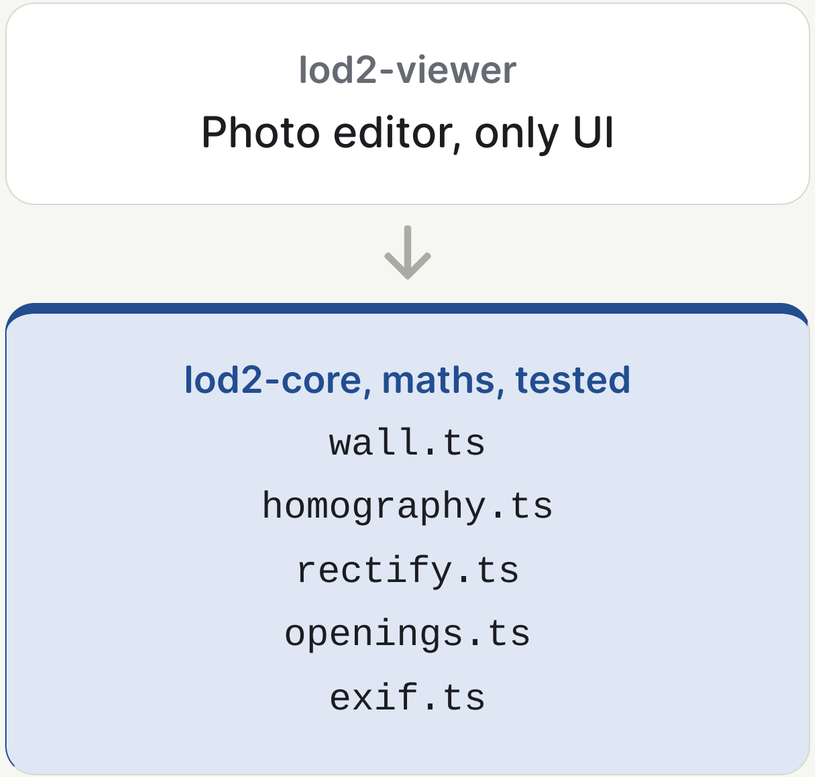
Lens
I did not correct lens distortion

All of this is written down in WRITEUP.md and MEASUREMENTS.md.

Code walkthrough

The route through the code

1. `wall.ts` the wall laid flat, and cut at the ground
2. `homography.ts` four pairs to a homography
3. `rectify.ts` the straight image, built backwards
4. `openings.ts` boxes in metres, the ratio
5. `exif.ts` only the compass
6. one test, then `PhotoEditor.tsx` as the caller



With more time

Next steps

1. Automatic openings, tested against my boxes
2. Mark openings behind cars as estimated
3. Save dots and openings per wall
4. Walls that are not rectangles
5. A printable report per wall

Thank you.

Questions?